Comparative Analysis of Student Academic Performance Prediction

This comparison examines the proposed student-performance prediction system alongside the research approaches reviewed for the project. The table compares the functions addressed by each approach; the charts present the same qualitative scope comparison.

1. Comparative Analysis Table

Model / Study	Core Approach	At-risk Prediction	Risk Score	Explainability	Fairness Audit	Intervention Guidance	What-if Analysis	Educator Workflow	Main Limitation / Gap
[1] Kaur (2025)	Decision Tree, Random Forest, SVM and Logistic Regression; feature selection.	Yes	No	Partial	No	Partial	No	No	Empirical dataset, validation and deployable intervention are not documented clearly.
[2] Yousafzai et al. (2021)	Attention-based BiLSTM with feature selection.	Partial	No	No	No	No	No	No	Grade prediction focus; no educator-facing risk score or intervention workflow described.
[3] Liu et al. (2022)	Evolutionary spiking neural network; membrane algorithm tunes model hyperparameters.	Partial	No	No	No	Partial	No	No	Predictive model study; explanations and student-facing actions are not its focus.
[4] Holicza & Kiss (2023)	SVM, Decision Tree, Random Forest and KNN.	Partial	No	Partial	No	No	No	No	Compares contexts and predictors; no integrated risk-management application described.
[5] Zhao et al. (2023)	Multiple ML algorithms with feature analysis.	No	No	Partial	No	No	No	No	Influencing-factor analysis is not a student-level explanation or intervention workflow.
[6] Liu et al. (2024) Auto-BRB	Automated Belief Rule Base with evidential reasoning; P-CMA-ES optimization and AIC.	Partial	No	Yes	No	No	No	No	Interpretable model contribution; risk scores, interventions and what-if use are not described.
[7] Kesgin et al. (2025)	Logistic Regression, Random Forest and XGBoost; SMOTE, 5-fold CV, SHAP and AIF360 fairness metrics.	Partial	No	Yes	Yes	No	No	No	Covers explainability and fairness; does not describe recommendations or an interactive what-if tool.
[8] El Jihaoui et al. (2026)	Nine calibrated models, threshold selection, SAGE, ALE, DiCE and intersectional fairness audit.	Yes	Yes	Yes	Yes	Partial	Yes	Partial	Most integrated comparator; evaluated on one regional cohort, not the project dataset.
[9] Vecchi (2026)	Comparative supervised ML, feature selection and interpretable-model analysis.	Yes	No	Yes	No	No	No	No	Focuses on model and feature behavior; calls for richer longitudinal / psychological data.
Proposed system	TabPFN on engineered tabular academic, behavioral and support-related features; risk score, explanations, recommendations and what-if.	Yes	Yes	Yes	No	Yes	Yes	Yes	Integration is the project contribution; results need reconciliation across the abstract and archived CSV, plus common-baseline and fairness evaluation.

Note: “Partial” indicates related or limited scope; “No” means the feature is not stated as a core function in the reviewed source. This is a qualitative comparison, not a ranking of predictive performance.

2. Graphical Comparative Analysis

The following graph compares whether each approach addresses seven decision-support functions. It is a qualitative scope comparison, not an accuracy, F1-score or user-study performance graph.

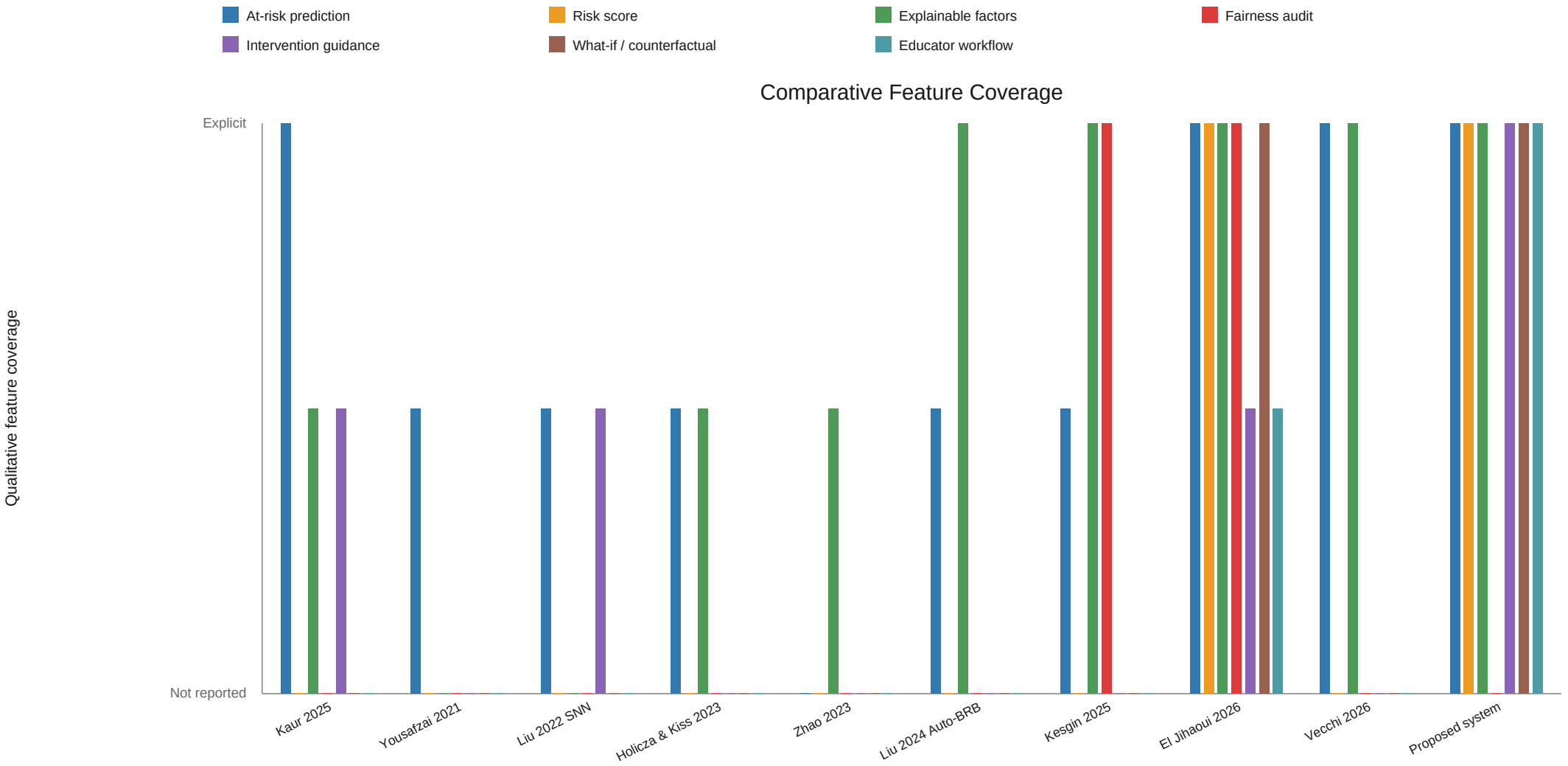

Figure 1. Qualitative feature coverage across the compared approaches. Full-height bars indicate explicit coverage; half-height bars indicate partial or aim-level coverage.

Figure 2 shows the breadth of feature areas addressed. Only explicit coverage is counted; partial coverage is excluded.

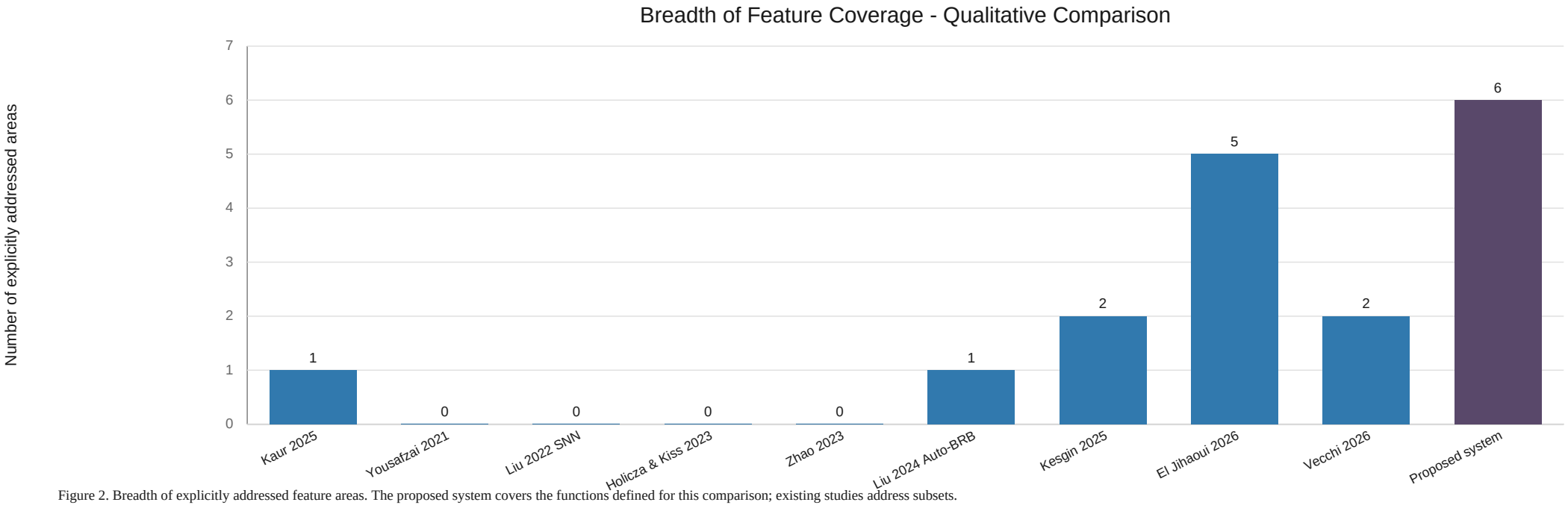

Figure 2. Breadth of explicitly addressed feature areas. The proposed system covers the functions defined for this comparison; existing studies address subsets.

3. Comparative Analysis / Research Gap

- Research spans grade prediction, performance classification and early-warning systems. The studies differ in population, outcome definition, predictors and validation, so reported metrics cannot be treated as a common ranking [1]-[9].
- Interpretability and fairness are already represented in the supplied literature: Auto-BRB uses interpretable rules, Kesgin et al. evaluate SHAP and subgroup fairness, and El Jihaoui et al. combine recall constraints, intersectional fairness and counterfactual explanations [6]-[8].
- The project opportunity is narrower: connect the UCI-based TabPFN prediction with a student-level risk score, understandable factors, intervention suggestions and what-if review in one educator-facing workflow.
- The UCI setting and final-grade-derived target require careful use. G3 must stay out of model inputs, and G1/G2 should only be used when they are available at the intended decision time. Subgroup fairness and external validation remain necessary.

- The proposed system addresses this application gap by combining TabPFN, the UCI Student Performance dataset and the requested risk-scoring, explanation, recommendation and what-if functions. Its contribution is this project-level integration; the comparison does not establish that TabPFN is more accurate than the models in other studies.

4. Conclusion of the Comparison

The reviewed work confirms that student performance can be modeled through classical machine learning, neural methods and interpretable rule systems. Recent studies also show that fairness and counterfactual explanations matter in early-warning settings. The proposed system brings prediction and educator-facing support functions together for its selected dataset. Its value should be judged through a common benchmark, transparent class-wise metrics, subgroup checks and educator review, rather than by comparing unrelated published scores.